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In [2]: import os
os.environ["LOKY_MAX_CPU_COUNT"] = "4"
os.environ["TF_CPP_MIN_LOG_LEVEL"] = "2"      # hides TensorFlow info messages

# Program 23: Cluster high-dimensional E-Commerce Customer Behavior data
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

FILE = r"D:\Datascience-MachineLearning\Datasets\ecommerce_customer_behavior_dataset.csv" # <-- change to your file path

df = pd.read_csv(FILE)
X = df.select_dtypes(include="number").dropna()
X_scaled = StandardScaler().fit_transform(X)

# choose the number of clusters using the silhouette score
best_k, best_score = 2, -1
for k in range(2, 7):
    labels = KMeans(n_clusters=k, random_state=42, n_init=10).fit_predict(X_scaled)
    score = silhouette_score(X_scaled, labels)
    print("k =", k, " silhouette =", round(score, 3))
    if score > best_score:
        best_k, best_score = k, score
print("\nBest k:", best_k)

labels = KMeans(n_clusters=best_k, random_state=42, n_init=10).fit_predict(X_scaled)
X = X.copy()
X["Cluster"] = labels

print("\nCustomers per cluster:")
print(X["Cluster"].value_counts())
print("\nAverage values of each cluster:")
print(X.groupby("Cluster").mean())

# visualise in 2D using PCA
pts = PCA(n_components=2).fit_transform(X_scaled)

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plt.scatter(pts[:, 0], pts[:, 1], c=labels, cmap="viridis", alpha=0.6)
plt.xlabel("PC 1")
plt.ylabel("PC 2")
plt.title("Customer Clusters")
plt.show()
```

```
k = 2  silhouette = 0.431
k = 3  silhouette = 0.14
k = 4  silhouette = 0.134
k = 5  silhouette = 0.13
k = 6  silhouette = 0.138
```

Best k: 2

Customers per cluster:

Cluster

1 4633

0 367

Name: count, dtype: int64

Average values of each cluster:

	Age	Unit_Price	Quantity	Discount_Amount	Total_Amount \
Cluster					
0	34.212534	2399.982044	2.757493	178.838038	5878.388174
1	35.097561	301.829741	2.177423	12.654967	595.332638

	Session_Duration_Minutes	Pages_Viewed	Delivery_Time_Days \
Cluster			
0	14.422343	8.940054	6.449591
1	14.585366	8.987697	6.500755

	Customer_Rating
Cluster	
0	3.934605
1	3.900281



In []: